

**Lab Report No _1_
Introduction to Edsim51 simulator with demo
examples
Embedded Systems**



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Lab Example No 1:

Solution:

Brief description (3-5 lines)

The program displays a binary pattern from 0 to 255 on LEDs connected to port 1. A 1 represents the LED on, while a 0 represents the LED off. To write the inverse of the pattern to the LEDs, the data FFH is sent to 0 and back to FFH. Since port 1 is initially at FFH, continuous decrement of port 1 is necessary to achieve this.

The code

start:

DEC P1 ; decrement port 1

JMP start ; and repeat

The results (Screenshot)



Lab Example No 2:

Solution:

Brief description (3-5 lines)

The program echoes P2 switches to P1 LEDs, with a closed switch triggering a logic 0 on the P2 pin, which is then copied to the P1 bit, turning on the LED, and vice versa.

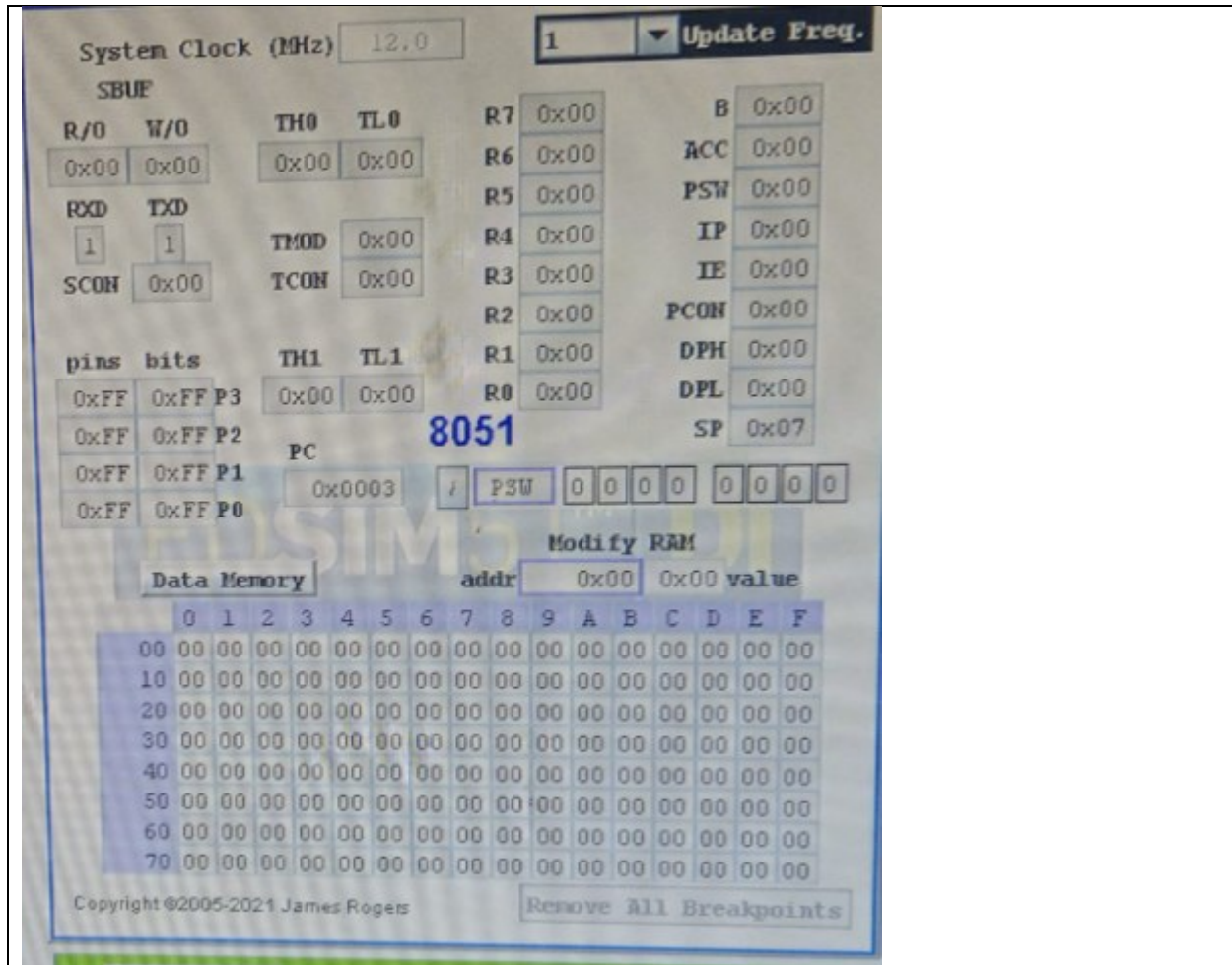
The code

start:

MOV P1, P2 ; move data on P2 pins to P1

JMP start ; and repeat

The results (Screenshot)



Lab Task No 1:

Solution:

Brief description (3-5 lines)

The 8051 can be used to interface an Analog-to-Digital Converter (ADC) to read an analog signal and display its digital equivalent.

The code

```
ORG 000H
MOV P1, #0FFH    ; Set Port 1 as input port (for ADC data)
MOV P2, #00H     ; Set Port 2 as output port (for control signals)
MOV DPTR, #LCD_CMD ; Initialize LCD Command Pointer
CALL INIT_LCD    ; Initialize the LCD
START:
```

```
SETB P2.0      ; Send high to start conversion (SC pin)
CLR P2.0       ; Send low to stop conversion
JB P2.1, START  ; Wait until end of conversion (EOC pin)
MOV A, P1       ; Read digital value from ADC (8-bit)
CALL DISPLAY    ; Send value to LCD for display
SJMP START      ; Repeat forever
```

DISPLAY:

; Code to display the value on LCD (implement your own LCD routine)

RET

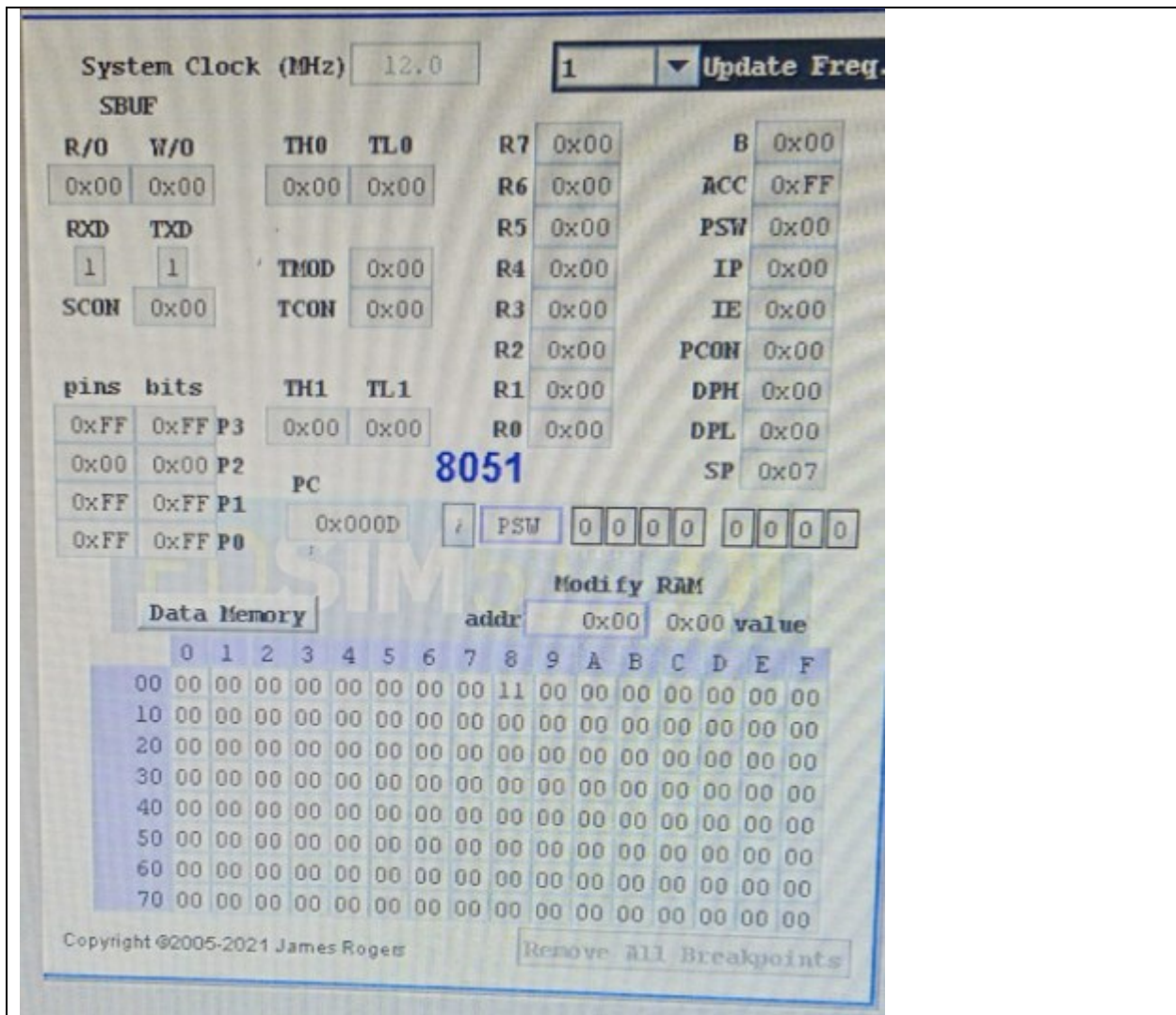
INIT_LCD:

; Code to initialize LCD (implement your own LCD routine)

RET

END

The results (Screenshot)



Conclusion

In conclusion, The Edsim51 simulator is a user-friendly tool for simulating the 8051 microcontrollers, offering an environment for assembly and C code development. It provides real-time debugging and code optimization, making it a valuable resource for both beginners and experienced developers.